Alpha-Seeker: A Text-Mining Pipeline for Alpha Discovery in Prediction Markets (Kalshi & Polymarket)

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Abstract

Prediction markets such as Kalshi and Polymarket have experienced significant growth in trading volume in recent years, making the rapid incorporation of public sentiment and breaking news into contract pricing increasingly important. This project proposes Alpha-Seeker, a text-based information retrieval and mining system designed to bridge the gap between noisy online discourse and actionable market insight. By retrieving and clustering real-time data from Reddit, the system identifies emerging themes and sentiment patterns related to macroeconomic and political contracts. Latent Dirichlet Allocation (LDA) will be used for unsupervised topic discovery alongside sentiment scoring techniques to structure the extracted information. The goal is to produce a signal-to-noise dashboard that enables analysts to detect emerging narratives and potential mispricings before they are reflected in market premiums.

Keywords

Information Retrieval, Text Mining, Topic Modeling, Prediction Markets, Sentiment Analysis

1 Introduction

Prediction markets differ from traditional financial markets in that they are driven by discrete real-world events, often influenced by fragmented and highly speculative information sources. Textual data from social platforms is abundant but noisy, unstructured, and rapidly evolving, creating challenges for extracting meaningful signals. This project addresses these challenges by designing a pipeline that filters low-value content, uncovers latent thematic relationships, and aligns textual sentiment with market movements. Filtering mechanisms will target spam and irrelevant discussion common in speculative communities, while topic modeling will identify discussions that may indirectly relate to contracts without explicit keyword references. Timestamped clustering outputs will be aligned with price series data from liquid markets accessed via the Kalshi API to examine potential lead-lag relationships between emerging narratives and contract price changes. The system therefore combines information retrieval, natural language processing, and financial data integration to explore whether structured textual analysis can reveal early indicators of market-relevant information.

2 Description of Work

The proposed system will be implemented as a Python-based pipeline composed of integrated modules to retrieve, analyze, and visualize market-relevant textual signals. The first component is a retrieval engine that gathers real-time discussions from online platforms using APIs and scraping tools. This module will use the PRAW library to collect Reddit data based on dynamically generated keywords such as Fed rate, CPI, Election, and plume detection. Collected text is then passed to a matching engine that identifies connections between extracted sentiment and tradeable prediction markets via the Kalshi API, aligning textual signals with market instruments.

The analytical core consists of a mining and clustering module designed to uncover latent structure within the corpus. Text data will be vectorized using TF-IDF, followed by LDA to cluster posts into coherent event-driven narratives. These clusters represent emerging discussions that may influence market expectations. Finally, an Alpha Dashboard built with Streamlit or Matplotlib presents the insights, displaying bullish and bearish sentiment indicators alongside current contract prices from Kalshi or Polymarket for an interpretable overview of potential information-driven signals.

The system will be evaluated at multiple levels to assess both technical quality and practical usefulness. Topic modeling performance will be measured using coherence scores (e.g., C_v) to evaluate how well related discussions are grouped into meaningful themes. A subset of posts may be manually labeled as relevant or irrelevant to selected contracts in order to estimate Precision and Recall of contract-related retrieval. Additionally, a lead-lag analysis will be conducted by comparing spikes in detected sentiment or topic intensity with subsequent movements in prediction market prices. This approach will help assess whether the identified signals meaningfully precede market changes.

The project is expected to be completed over four main phases. The first phase will focus on API setup and data ingestion from Reddit and prediction market platforms. The second phase will involve preprocessing and implementation of the topic modeling framework. The third phase will center on sentiment scoring and development of a simple visualization dashboard. The final phase will involve system evaluation, backtesting analysis, and preparation of the final report and presentation. Responsibilities will be distributed across team members by component, with collaborative integration and evaluation toward the end of the project timeline.